

$$\begin{aligned}
& \frac{1}{16} \tilde{m}_2 \tilde{\mu} g_2^4 \left(\frac{1}{3} m_2 \tilde{\mu} g_2^4 (-3 y_e^{i1i2} + 2 y_e^{pi2} \delta_{pi1}) \text{LF}_{2,2,0}[m_2, \tilde{\mu}] + \right. \\
& \frac{1}{6} m_2 \tilde{\mu} g_2^4 (3 y_e^{i1i2} - 2 y_e^{pi2} \delta_{pi1}) \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] - \\
& \frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_e^{i1i2} \overline{a_d^{pr}} \text{LF}_{2,2,0}[m_d^r, m_q^p] - \frac{1}{4} \tilde{\mu} g_1^2 y_e^{pr} y_e^{i1i2} \overline{a_e^{pr}} \text{LF}_{2,2,0}[m_e^r, m_l^p] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_e^{pr} y_e^{i1i2} \overline{a_e^{pr}} \text{LF}_{3,1,0}[m_l^p, m_e^r] + \frac{1}{4} \tilde{\mu} g_1^2 y_e^{pr} y_e^{i1i2} \overline{a_e^{pr}} \text{LF}_{3,2,-1}[m_l^p, m_e^r] - \\
& \frac{1}{8} \tilde{\mu} y_e^{pr} \overline{a_e^{pr}} (y_e^{i1i2} (9 g_1^2 + g_2^2) - 2 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{4,1,-1}[m_l^p, m_e^r] + \\
& \frac{1}{6} \tilde{\mu} y_e^{pr} \overline{a_e^{pr}} (3 g_1^2 y_e^{i1i2} - 2 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{5,1,-2}[m_l^p, m_e^r] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_d^{pr} y_e^{i1i2} \overline{a_d^{pr}} \text{LF}_{3,1,0}[m_q^p, m_d^r] + \frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_e^{i1i2} \overline{a_d^{pr}} \text{LF}_{3,2,-1}[m_q^p, m_d^r] - \\
& \frac{3}{8} \tilde{\mu} y_d^{pr} \overline{a_d^{pr}} (y_e^{i1i2} (5 g_1^2 + g_2^2) - 2 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{4,1,-1}[m_q^p, m_d^r] + \\
& \frac{1}{2} \tilde{\mu} y_d^{pr} \overline{a_d^{pr}} (3 g_1^2 y_e^{i1i2} - 2 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{5,1,-2}[m_q^p, m_d^r] - \\
& \frac{1}{8} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (y_e^{i1i2} (g_1^2 + 9 g_2^2) - 4 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{2,2,0}[m_q^p, m_u^r] + \\
& \frac{3}{4} \tilde{\mu} g_2^2 y_e^{i1i2} y_u^{pr} \overline{a_u^{pr}} \text{LF}_{3,1,0}[m_q^p, m_u^r] - \frac{1}{4} \tilde{\mu} g_2^2 y_e^{si2} y_u^{pr} \overline{a_u^{pr}} \text{LF}_{3,2,-1}[m_q^p, m_u^r] \delta_{si1} - \\
& \frac{3}{4} \tilde{\mu} g_2^2 y_e^{i1i2} y_u^{pr} \overline{a_u^{pr}} \text{LF}_{4,1,-1}[m_q^p, m_u^r] + \\
& \frac{1}{4} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (7 g_1^2 y_e^{i1i2} - 2 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{3,1,0}[m_u^r, m_q^p] + \\
& \frac{1}{8} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (y_e^{i1i2} (g_1^2 + 9 g_2^2) - 4 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{3,2,-1}[m_u^r, m_q^p] + \\
& \frac{3}{4} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (-5 g_1^2 y_e^{i1i2} + 2 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{4,1,-1}[m_u^r, m_q^p] + \\
& \frac{1}{2} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (3 g_1^2 y_e^{i1i2} - 2 g_2^2 y_e^{si2} \delta_{si1}) \text{LF}_{5,1,-2}[m_u^r, m_q^p] + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} (g_1^2 - g_2^2) \text{LF}_{3,1,0}[\tilde{\mu}, m_1] + \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{8} m_1 \tilde{\mu} g_1^2 (3 y_e^{i1i2} (-2 g_1^2 + g_2^2) + 2 g_2^2 y_e^{pi2} \delta_{pi1}) \text{LF}_{4,1,-1}[\tilde{\mu}, m_1] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 (3 g_1^2 y_e^{i1i2} - 2 g_2^2 y_e^{pi2} \delta_{pi1}) \text{LF}_{5,1,-2}[\tilde{\mu}, m_1] + \\
& \frac{1}{12} m_2 \tilde{\mu} g_2^2 (3 y_e^{i1i2} (3 g_1^2 + g_2^2) - 8 g_2^2 y_e^{pi2} \delta_{pi1}) \text{LF}_{3,1,0}[\tilde{\mu}, m_2] + \\
& \frac{1}{3} m_2 \tilde{\mu} g_2^4 (9 y_e^{i1i2} - 2 y_e^{pi2} \delta_{pi1}) \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\
& \frac{1}{8} m_2 \tilde{\mu} g_2^2 (-3 y_e^{i1i2} (6 g_1^2 + g_2^2) + 14 g_2^2 y_e^{pi2} \delta_{pi1}) \text{LF}_{4,1,-1}[\tilde{\mu}, m_2] - \\
& 2 m_2 \tilde{\mu} g_2^4 y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{2} m_2 \tilde{\mu} g_2^2 (3 g_1^2 y_e^{i1i2} - 2 g_2^2 y_e^{pi2} \delta_{pi1}) \text{LF}_{5,1,-2}[\tilde{\mu}, m_2] - \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,0}[m_1, m_e^r, m_l^{i1}] - \\
& \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,1,-1}[m_1, m_e^r, m_l^{i1}] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \\
& \text{LF}_{3,1,1,-1}[m_1, m_e^r, m_l^{i1}] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,0}[m_1, m_e^r, \tilde{\mu}] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-1}[m_1, m_e^r, \tilde{\mu}] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} (-g_1^2 + g_2^2) \\
& \text{LF}_{2,2,1,-1}[m_1, m_e^{i2}, m_l^{i1}] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,1,-1}[m_1, m_l^{i1}, m_e^r] + \\
& \frac{1}{16} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} (-g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_1, m_l^{i1}, m_e^{i2}] - \frac{3}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \\
& \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^r] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} (-9 g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^{i2}] + \\
& \frac{1}{16} m_1 \tilde{\mu} g_1^2 (2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} (9 g_1^2 - g_2^2)) \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{i1}] + \\
& \frac{3}{4} \tilde{\mu} g_1^2 g_2^2 y_e^{i1i2} (3 m_1 + m_2) \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& \tilde{\mu} g_1^2 g_2^2 y_e^{i1i2} (m_1 + m_2) \text{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
& \frac{1}{8} m_2 \tilde{\mu} g_2^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_e^r] - \frac{1}{2} m_2 \tilde{\mu} g_2^4 y_e^{pi2} \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^p] \delta_{pi1} - \\
& \frac{1}{16} m_2 \tilde{\mu} g_2^2 (6 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} (3 g_1^2 + 13 g_2^2)) \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{i1}] + \\
& \frac{3}{4} \tilde{\mu} \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_e^{i1i2} \overline{a_d^{pr}} \text{LF}_{2,2,1,-1}[m_d^r, m_q^p, m_d^t] - \\
& \frac{3}{4} \tilde{\mu} \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_e^{i1i2} \overline{a_d^{pr}} \text{LF}_{2,2,1,-1}[m_d^t, m_q^p, m_d^r] + \\
& \frac{1}{4} \tilde{\mu} \overline{y_e^{st}} y_e^{pt} y_e^{sr} y_e^{i1i2} \overline{a_e^{pr}} \text{LF}_{2,2,1,-1}[m_e^r, m_l^p, m_e^t] - \\
& \frac{1}{4} \tilde{\mu} \overline{y_e^{st}} y_e^{pt} y_e^{sr} y_e^{i1i2} \overline{a_e^{pr}} \text{LF}_{2,2,1,-1}[m_e^t, m_l^p, m_e^r] - \\
& \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} (3 g_1^2 + g_2^2) \text{LF}_{2,1,1,0}[m_e^{i2}, m_1, m_l^{i1}] + \\
& \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_e^{i2}, m_1, m_l^{i1}] + \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{2,1,1,0}[m_e^{i2}, m_1, \$$